

Real-Time PPE Compliance Monitoring System

Automated Safety Surveillance for Industrial & Construction Sites

Architecture, Model & Data Pipeline

Objective: Develop an automated, real-time computer vision system to monitor Personal Protective Equipment (PPE) compliance, mitigate workplace accidents, and maintain strict industrial safety standards.

Core Functionality: Automatic detection of safety gear violations (hard hats, masks, safety vests) alongside machinery and personnel tracking across live camera feeds and recorded videos.

Review Focus Areas:

- Construction Site Safety Dataset & Class Distribution
- YOLOv8 Computer Vision Architecture & Training
- End-to-End System & API Architecture
- Complete Technology Stack
- Expected Analytical Outputs & Incident Logging

Project Overview & Problem Statement

Mitigating Industrial Hazards through Computer Vision

The Industrial Challenge

- Construction and manufacturing zones face high occupational hazards and safety non-compliance.
- Manual safety inspection is labour-intensive, error-prone, inconsistent, and cannot provide continuous 24/7 coverage.

Proposed Solution

- Real-time automated visual inspection pipeline using state-of-the-art object detection.
- Instantaneous identification of compliant vs. non-compliant personnel.
- Automated generation of violation logs with timestamped visual evidence.

Key Monitored Classes

Compliance Classes

Hardhat, Mask, Safety Vest

Violation Classes

NO-Hardhat, NO-Mask, NO-Safety Vest

Contextual & Neutral Classes

Person, Machinery, Vehicle, Safety Cone

Dataset Characteristics & Class Distribution

Construction Site Safety Dataset (Kaggle)

Dataset Source & Link

Kaggle URL:

<https://www.kaggle.com/datasets/snehilsanyal/construction-site-safety-image-dataset-roboflow>

Origin: Construction Site Safety Image Dataset (Roboflow Universe via Kaggle)

Dataset Scale & Partition

- Total Images: 2,801 annotated images
- Total Annotations: 38,352 object bounding boxes
- Training Split: 2,240 images (30,684 annotations, ~80%)
- Validation Split: 280 images (3,554 annotations, ~10%)
- Test Split: 281 images (4,114 annotations, ~10%)

Class Annotation Breakdown

- Person: 9,872 instances (primary contextual object)
- Machinery & Vehicles: 5,346 machinery and 1,628 vehicle instances
- Safety Equipment: 4,158 NO-Safety Vest, 3,502 Safety Cone, 3,334 Hardhat, 3,250 NO-Mask, 3,135 Safety Vest, 2,427 NO-Hardhat, 1,700 Mask

Dataset Relevance

High diversity of angles, lighting conditions, occlusions, and multi-worker density typical of live industrial sites.

Data Augmentation & Preprocessing Pipeline

Robustness Engineering for Dynamic Industrial Environments

Input Preprocessing

- **Spatial Standardisation:** 640 x 640 pixel resolution (BCHW tensor format)
- **Normalisation:** Pixel values scaled to [0.0, 1.0] range

Data Augmentation Strategy

- **Photometric Distortions:** HSV colour jittering (Hue: 0.015, Saturation: 0.7, Value/Brightness: 0.4) for varying daylight conditions
- **Geometric Transformations:** Rotation (+-5.0 degrees), Translation (+-0.1), Scaling (+-50%), Shear (+-2.0 degrees), Horizontal Flip (p = 0.5)
- **Advanced Composition:** Mosaic Augmentation (scale 1.0) and Mixup (0.1) to boost detection of small and partially occluded safety gear
- **Regularisation:** Copy-Paste augmentation (0.1) and disabling Mosaic in the final 10 epochs for fine-grained boundary stabilisation

Computer Vision Model Architecture

YOLOv8m (Ultralytics) Single-Stage Object Detector

Model Selection: YOLOv8m (Medium): Balances high inference accuracy (mAP) with real-time frame rates (>40 FPS).

Backbone

Modified CSPDarknet53 with C2f (Cross-Stage Partial with 2 Convolutions) modules for rich multi-scale feature extraction.

Neck

Path Aggregation Network (PANet) and Feature Pyramid Network (FPN) for seamless multi-scale contextual semantic fusion.

Detection Head

Anchor-free, decoupled head separating classification and bounding box regression branches to accelerate convergence.

Loss Functions

Complete IoU (CIoU) and Distribution Focal Loss (DFL) for precise bounding box localisation. Binary Cross-Entropy (BCE) with Label Smoothing (0.1) for class classification.

Model Training, Hardware & Performance Metrics

Rigorous Training Pipeline & Quantitative Evaluation

Training Infrastructure

- Hardware: Dual NVIDIA Tesla T4 GPUs (16 GB VRAM each)
- Framework: PyTorch 2.10.0, CUDA 13.0, Ultralytics 8.4.30
- Optimization: SGD (momentum: 0.937, weight decay: 0.0005) with Cosine Annealing Learning Rate Schedule (Initial LR: 0.01, Final LR: 0.0001)
- Training Duration: 100 Epochs (Best model achieved at Epoch 88)

Quantitative Results (Test Set Evaluation)

Overall Performance: mAP@0.5 = 0.8807 (88.07%),
mAP@0.5:0.95 = 0.6425

Precision & Recall: Precision = 92.27%, Recall = 80.92%

Critical Class Accuracies (AP@0.5): Machinery (96.40%), Mask (95.48%), Person (93.38%), Vehicle (90.57%), NO-Safety Vest (89.17%), NO-Hardhat (87.62%), Safety Vest (85.05%), NO-Mask (82.98%), Hardhat (82.76%)

Inference Speed: 23.5 ms per frame (42 FPS), fully supporting real-time streaming

System Architecture & Workflow Pipeline

Client-Server Decoupled Architecture with Asynchronous Inference

End-to-End Operational Pipeline

01

Input Layer

User uploads live feed, single image, or batch video stream via modern web client.

03

Inference Engine (YOLOv8 Core)

Executes forward pass, applies Non-Maximum Suppression (NMS), and classifies 10 target classes.

05

Persistence & Analytics Store

Writes incident frames, bounding box coordinates, and session analytics to local JSON datastore.

02

API Gateway (FastAPI)

Validates media payloads, enforces size limits, and delegates video tasks to an asynchronous background worker pool.

04

Temporal Smoothing Filter

5-frame temporal buffer prevents transient false alarms caused by temporary occlusions.

06

Presentation Layer

Delivers real-time annotated visual overlays, live safety metrics, and alert triggers.

Tools & Technologies Stack

End-to-End Modern Software & ML Stack

Machine Learning & Vision

- YOLOv8 (Ultralytics)
- PyTorch
- ONNX Runtime (Opset 17 export for edge acceleration)
- OpenCV
- NumPy

Backend & Service Layer

- FastAPI (Python 3.10+ high-performance asynchronous REST API)
- Uvicorn (ASGI server implementation)
- Pydantic (data validation and settings management)
- BackgroundTasks (non-blocking asynchronous video decoding and processing)

Frontend & User Interface

- React 18 with TypeScript
- Vite 5 (high-speed build tool and development server)
- Tailwind CSS 3.4 (utility-first responsive styling)
- Radix UI & shadcn/ui (accessible UI component primitives)
- Framer Motion (interactive micro-animations and dashboard transitions)
- Lucide React (vector iconography)

Expected Outputs & Application Deliverables

Comprehensive Safety Monitoring & Incident Analytics

Visual Detections

Real-time bounding box annotations with distinct colour-coding (Green for Compliance, Red for Violations, Cyan for Site Context).

Interactive Safety Dashboard

Real-time Site Safety Compliance Score (percentage of compliant vs total workers). Detection Metrics: Live mAP indicator, active alert frequency, and processed frame rates.

Automated Incident Logging

Immediate snapshot capture of violation events.
Timestamped tabular log with worker identifier, violation category, and confidence score.

Operational Control & Customisation

Dynamic confidence threshold (0.1 to 0.95) and NMS IoU adjustment from the UI. Model weight hot-reloading capability for rapid updates without server restarts.

Summary, Progress Status & Future Roadmap

Review #1 Milestones & Next Phase Execution

Milestones Completed (Review #1)

- Curation and preprocessing of 2,801 construction safety images across 10 classes.
- Full training and evaluation of YOLOv8m model achieving 88.07% mAP@0.5.
- Implementation of FastAPI asynchronous backend and React TypeScript UI dashboard.
- End-to-end integration of single image and multi-frame video inference pipelines.

Next Steps (Review #2 & Beyond)

- Integration of real-time RTSP camera streaming protocols.
- Multi-camera zone management and automated alert notifications (SMS/Email/WebSockets).
- Model quantisation (INT8 / TensorRT) for deployment on edge devices like NVIDIA Jetson.